```java
import java.io.*;

public class FIFO_PageReplacement {
  public static void main(String[] args) throws IOException {

    BufferedReader reader = new BufferedReader(new InputStreamReader(System.in));

    // Variables
    int totalFrames;         // Number of available memory frames
    int nextReplaceIndex = 0; // Points to the frame that will be replaced next
    int hitCount = 0;        // Number of page hits
    int faultCount = 0;      // Number of page faults
    int refStringLength;     // Length of reference string

    int[] frames;            // Memory frames
    int[] referenceString;    // Reference string (sequence of page requests)
    int[][] memoryLayout;     // To store memory status after each page request

    // Step 1: Input number of frames
    System.out.print("Enter the number of frames: ");
    totalFrames = Integer.parseInt(reader.readLine());

    // Step 2: Input length of reference string
    System.out.print("Enter the length of the reference string: ");
    refStringLength = Integer.parseInt(reader.readLine());

    // Step 3: Initialize arrays
    frames = new int[totalFrames];
    referenceString = new int[refStringLength];
    memoryLayout = new int[refStringLength][totalFrames];
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// Step 4: Set all frames as empty (-1 means empty)
for (int i = 0; i < totalFrames; i++) {
    frames[i] = -1;
}

// Step 5: Input the reference string
System.out.println("Enter the reference string (space-separated or line-by-line): ");
for (int i = 0; i < refStringLength; i++) {
    referenceString[i] = Integer.parseInt(reader.readLine());
}

System.out.println();

// Step 6: Process each page in the reference string
for (int i = 0; i < refStringLength; i++) {
    int currentPage = referenceString[i];
    boolean pageFound = false;

    // Step 6.1: Check if the page is already in a frame (HIT)
    for (int j = 0; j < totalFrames; j++) {
        if (frames[j] == currentPage) {
            hitCount++;
            pageFound = true;
            break;
        }
    }

    // Step 6.2: If page not found (FAULT), replace using FIFO
    if (!pageFound) {
        frames[nextReplaceIndex] = currentPage;
        faultCount++;
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            nextReplaceIndex++;

            // Wrap around if we reach the end of the frame list
            if (nextReplaceIndex == totalFrames) {
                nextReplaceIndex = 0;
            }
        }

        // Step 6.3: Record current frame status for display later
        for (int j = 0; j < totalFrames; j++) {
            memoryLayout[i][j] = frames[j];
        }
    }

    // Step 7: Display memory layout after processing all pages
    System.out.println("Memory layout after each reference:");
    for (int frame = 0; frame < totalFrames; frame++) {
        for (int ref = 0; ref < refStringLength; ref++) {
            System.out.printf("%3d ", memoryLayout[ref][frame]);
        }
        System.out.println();
    }

    // Step 8: Display hit/fault summary
    System.out.println("\nTotal Page Hits  : " + hitCount);
    System.out.println("Total Page Faults: " + faultCount);
    System.out.printf("Hit Ratio        : %.2f\n", (float) hitCount / refStringLength);
  }
}
```

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import java.io.*;
public class FIFO {
public static void main(String[] args) throws IOException
{
BufferedReader br = new BufferedReader(new
InputStreamReader(System.in));
int frames, pointer = 0, hit = 0, fault = 0,ref_len;
int buffer[];
int reference[];
int mem_layout[][];
System.out.println("Please enter the number of Frames: ");
frames = Integer.parseInt(br.readLine());
System.out.println("Please enter the length of the Reference string: ");
ref_len = Integer.parseInt(br.readLine());
reference = new int[ref_len];
mem_layout = new int[ref_len][frames];
buffer = new int[frames];
for(int j = 0; j < frames; j++)
buffer[j] = -1;
System.out.println("Please enter the reference string: ");
for(int i = 0; i < ref_len; i++)
{
reference[i] = Integer.parseInt(br.readLine());
}
System.out.println();
for(int i = 0; i < ref_len; i++)
{
int search = -1;
for(int j = 0; j < frames; j++)
{
if(buffer[j] == reference[i])
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{
search = j;
hit++;
break;
}
}
if(search == -1)
{
buffer[pointer] = reference[i];
fault++;
pointer++;
if(pointer == frames)
pointer = 0;
}
for(int j = 0; j < frames; j++)
mem_layout[i][j] = buffer[j];
}
for(int i = 0; i < frames; i++)
{
for(int j = 0; j < ref_len; j++)
System.out.printf("%3d ",mem_layout[j][i]);
System.out.println();
}
System.out.println("The number of Hits: " + hit);
System.out.println("Hit Ratio: " + (float)((float)hit/ref_len));
System.out.println("The number of Faults: " + fault);
}
}
```